

D653

Empathizing, Defining, and Ideating

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A team at a regional transit agency is redesigning its fare app. The lead designer, who has won two industry awards, says the team rides the bus daily and already knows what riders need, so interviews would waste the six-week timeline. What is the strongest response in a design thinking process?

- A. Skip research only if the timeline is shorter than eight weeks.
- B. Run short interviews and observations with a range of riders before defining the problem.
- C. Proceed with the team's insights, since they are daily riders.
- D. Defer to the lead designer because of her award record.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

After interviewing twenty home health aides, a team at a staffing company with offices in four states finds that aides lose time between visits because schedules change without notice. Which problem statement best fits a user-centered define stage?

- A. The company needs to reduce scheduling costs across all four state offices.
- B. Home health aides need a reliable way to know about schedule changes before they travel, because surprise changes waste time between visits.
- C. Home health aides need a mobile app with push notifications for schedule changes.
- D. Aides are frustrated.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3. [TRANSFER — reversed stem / different variable]

A team's problem statement reads: "Busy parents need a meal-planning chatbot." What is the main flaw?

- A. It names a specific user group.
- B. It is too short; problem statements must be at least three sentences.
- C. It should describe the company's revenue goal.
- D. It prescribes a solution (a chatbot) instead of stating the parents' need and insight.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

A design team at a credit union with 40,000 members drafts a persona: “Maria, 34, earns \$62,000, likes yoga and coffee.” The team has twelve interview transcripts and a survey on hand. What should the team add to make the persona useful?

- A. A note that she represents all 40,000 members.
- B. More demographic detail such as zip code and education level.
- C. A photo and a quote invented to make her feel real.
- D. Goals, behaviors, and frustrations drawn from patterns in the interviews and survey.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5. [TRANSFER — reversed stem / different variable]

A team has a persona with clear goals and pain points, but no one on the team can say where those details came from. What is the risk?

- A. The persona will be too data-heavy for stakeholders.
- B. The persona may reflect the team’s assumptions rather than real users, steering design in the wrong direction.
- C. The persona will need to be turned into a journey map.
- D. There is no risk as long as the persona has goals and pain points.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

Eight people from design, engineering, and customer support meet for a 45-minute ideation session in a newly renovated room. Ten minutes in, they have three ideas because an engineer points out a technical limit for each one as it is raised. What should the facilitator do?

- A. Remove the engineer so the session is design-only.
- B. Extend the session to 90 minutes.
- C. Keep the feasibility checks so only buildable ideas move forward.
- D. Ask the group to defer judgment and aim for quantity, saving feasibility review for a later converge step.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7. [TRANSFER — reversed stem / different variable]

A team has 60 sticky-note ideas on the wall. What should happen next?

- A.** Group similar ideas and narrow them using criteria tied to user needs, such as impact and feasibility.
- B.** Go back and rewrite the persona.
- C.** Keep generating ideas until there are at least 100.
- D.** Build all 60 as prototypes.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

A researcher at a pet-supply startup that just closed a funding round is preparing questions for dog owners. The product team hopes owners want an automatic feeder. Which question best follows user research practice?

- A.** “Do you feed your dog?”
- B.** “Wouldn’t an automatic feeder make your life easier?”
- C.** “Tell me about the last time you had to arrange feeding for your dog while you were away.”
- D.** “Would you buy an automatic feeder for \$80?”

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9. [TRANSFER — reversed stem / different variable]

Interview notes show every participant agreed a proposed feature was “a great idea,” but usage data later shows almost no one used it. What is the most likely research flaw?

- A.** The users were lying and research cannot be trusted.
- B.** The sample was too large.
- C.** Questions were leading or hypothetical, so participants gave polite agreement instead of describing real behavior.
- D.** The team skipped the ideation stage.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

During a two-hour field visit at a busy clinic with new flooring, a researcher watches a nurse print a patient list, cross out names by hand, and re-enter them into the system. Where does this observation belong on an empathy map?

- A.** Feels
- B.** Says
- C.** Does
- D.** It belongs on the feature backlog, not the empathy map.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A city library's new booking site works well in testing with staff and college volunteers. A screen-reader user and an older adult with limited broadband both failed to finish a booking. The project sponsor, a longtime council member, calls them edge cases. What is the best design response?

- A.** Include these users in research and fix the barriers, since solving for them usually improves the site for everyone.
- B.** Accept the sponsor's view because of their role on the council.
- C.** Edge cases should never shape a public product.
- D.** Launch now and address accessibility in a later phase.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

A hotel chain with 90 locations wants to learn where guests grow frustrated from the moment they book until checkout. The team has survey quotes and call-center logs. Which artifact best shows this?

- A.** A user journey map covering each stage from booking to checkout with actions, emotions, and pain points.
- B.** A list of all 90 locations ranked by rating.
- C.** A site map of the booking website.
- D.** A persona of the typical guest.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer B Concept 1

A team at a regional transit agency is redesigning its fare app. The lead designer, ~~who has won two industry awards~~, says the team rides the bus daily and already knows what riders need, so interviews would waste the ~~six-week~~ timeline. What is the strongest response in a design thinking process?

Target word: what the empathize stage requires **Decides:** Team members are not the users; empathy needs evidence from real riders.

A	Wrong stem number	Grabs the timeline as the deciding fact; timeline changes the depth of research, not whether to do it.
B	Correct	Empathize gathers evidence from actual users; the team's own habits are assumptions to test.
C	Adjacent category	Treats the design team as a proxy for users, which skips the empathize stage.
D	Camouflage	Awards do not replace user research.

2. Answer B Concept 2

After interviewing twenty home health aides, a team at a staffing company ~~with offices in four states~~ finds that aides lose time between visits because schedules change without notice. Which problem statement best fits a user-centered define stage?

Target word: best user-centered problem statement **Decides:** Names user, need, and insight without naming a solution.

A	Camouflage	Reframes around the business and the office count, dropping the user.
B	Correct	Specific user, a need, and the research insight; no solution is prescribed.
C	Inverted operation	Starts from a solution and works backward; the problem statement should leave room for ideation.
D	Dropped element	Names a user and a feeling but leaves out the need and the insight.

3. Answer D Concept 2 · transfer

A team's problem statement reads: "Busy parents need a meal-planning chatbot." What is the main flaw?

Target word: diagnose a flawed problem statement **Decides:** It names a solution instead of a need.

A	Inverted operation	Naming a specific user is a strength, not a flaw.
B	Absolute language	Length is not the test; the content is.
C	Adjacent category	Business goals are a separate input; the user problem statement centers the user.
D	Correct	A solution in the problem statement narrows ideation before it starts.

4. Answer D Concept 3

A design team at a credit union ~~with 40,000 members~~ drafts a persona: "Maria, 34, earns \$62,000, likes yoga and coffee." The team has twelve interview transcripts and a survey on hand. What should the team add to make the persona useful?

Target word: what a research-based persona needs **Decides:** Personas need goals, behaviors, and pain points drawn from research.

A	Absolute language	One persona represents a segment, not every member.
B	Adjacent category	More demographics still do not tell designers what the user is trying to do.
C	Dropped element	Adds polish but still skips the research-based goals and pain points.
D	Correct	A persona is useful when it reflects real user goals and pain points, not only demographics.

5. Answer B Concept 3 • transfer

A team has a persona with clear goals and pain points, but no one on the team can say where those details came from. What is the risk?

Target word: risk of a persona without research **Decides:** Unsupported personas encode assumptions.

A	Inverted operation	The problem is too little data, not too much.
B	Correct	Personas are only as valid as the research behind them.
C	Confusion set	Journey maps are a different tool and do not fix missing evidence.
D	Absolute language	Having the right fields does not make the content true.

6. Answer D Concept 4

Eight people from design, engineering, and customer support meet for a 45-minute ideation session ~~in a newly renovated room~~. Ten minutes in, they have three ideas because an engineer points out a technical limit for each one as it is raised. What should the facilitator do?

Target word: fix a stalled ideation session **Decides:** Judgment during divergence suppresses idea volume.

A	Adjacent category	Interdisciplinary input is valuable; the fix is the rule, not the roster.
B	Wrong stem number	More time does not fix the habit that is killing ideas.
C	Inverted operation	Applies converging criteria during the diverging step.
D	Correct	Divergent thinking comes first; evaluation comes after many ideas exist.

7. Answer A Concept 4 • transfer

A team has 60 sticky-note ideas on the wall. What should happen next?

Target word: the converge step **Decides:** After divergence, cluster and select with criteria.

A	Correct	Converging narrows many ideas with clear criteria.
B	Confusion set	Personas belong to define; nothing suggests the persona is wrong.
C	Absolute language	There is no magic count; with a large set in hand, the process moves to converging.
D	Dropped element	Skips selection entirely.

8. Answer C Concept 5

A researcher at a pet-supply startup ~~that just closed a funding round~~ is preparing questions for dog owners. The product team hopes owners want an automatic feeder. Which question best follows user research practice?

Target word: best interview question **Decides:** Open-ended, about past behavior, not leading.

A	Dropped element	Yes/no; produces no story or insight.
B	Inverted operation	Leads the user toward the answer the team wants.
C	Correct	Open, grounded in a real past event, and does not suggest a solution.
D	Adjacent category	A pricing and intent question, not a question about behavior and needs; stated intent is unreliable.

9. Answer C Concept 5 • transfer

Interview notes show every participant agreed a proposed feature was “a great idea,” but usage data later shows almost no one used it. What is the most likely research flaw?

Target word: diagnose research flaw from the outcome **Decides:** Leading/hypothetical questions produce polite agreement.

A	Absolute language	Overgeneralizes; the method, not research itself, failed.
B	Inverted operation	A larger sample is not what produces uniform agreement.
C	Correct	Asking about opinions of a proposed idea invites agreement; past behavior predicts use better.
D	Confusion set	Ideation is not a research-method issue.

10. Answer C Concept 6

During a ~~two-hour~~ field visit at a busy clinic ~~with new flooring~~, a researcher watches a nurse print a patient list, cross out names by hand, and re-enter them into the system. Where does this observation belong on an empathy map?

Target word: empathy map quadrant **Decides:** An observed action goes in Does.

A	Confusion set	Feelings may be inferred later, but the observation itself is an action.
B	Confusion set	Says is for direct quotes; no words were recorded.
C	Correct	The researcher observed an action.
D	Adjacent category	An empathy map organizes research; jumping to features skips define.

11. Answer A Concept 7

A city library’s new booking site works well in testing with staff and college volunteers. A screen-reader user and an older adult with limited broadband both failed to finish a booking. The project sponsor, ~~a longtime council member~~, calls them edge cases. What is the best design response?

Target word: treatment of excluded users **Decides:** Inclusive design treats excluded users as a source of better design for all.

A	Correct	Equitable design centers those most often excluded; fixes like clear structure and light pages help all users.
B	Camouflage	The sponsor’s title does not settle a design question.
C	Absolute language	The opposite holds for public services, which must work for all residents.
D	Adjacent category	Treats access as an add-on rather than part of core design.

12. Answer A Concept 8

A hotel chain ~~with 90 locations~~ wants to learn where guests grow frustrated from the moment they book until checkout. The team has survey quotes and call-center logs. Which artifact best shows this?

Target word: which artifact shows frustration across steps **Decides:** A journey map follows steps and emotions over time.

A	Correct	Journey maps show where, across a sequence, the experience breaks down.
B	Camouflage	Uses the location count; ratings do not show which step causes frustration.
C	Confusion set	Shows page structure, not the guest's experience across stages.
D	Adjacent category	Describes who the guest is, not where the journey fails.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	4	
Wrong stem number	A figure copied from the prompt that is not the answer.	2	
Adjacent category	The right idea in the wrong bucket.	8	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	6	
Absolute language	Always, never, guarantee — where judgment applies.	6	
Confusion set	Every option is real; the test is exact role-matching.	6	
Camouflage	Answers a stem fact that was never the question.	4	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Empathy comes from research, not assumption	The empathize stage gathers evidence about real users' experiences through observation and interviews before any solution is chosen.	Designing from the team's own preferences or a stakeholder's hunch and calling it user-centered.	A team that wants to skip talking to users because it "already knows" what they need.
2	User-centered problem statements	A problem statement names a specific user, their need, and the insight behind it, without prescribing a solution.	Writing the solution into the problem ("Users need an app that...").	A choice between framings where one already names a feature.
3	Personas are built from research data	A persona is a composite of patterns observed across real users: goals, behaviors, frustrations, and context.	Building personas from demographics or stereotypes alone.	A persona draft full of age, income, and hobbies but no goals or pain points.
4	Diverge, then converge	Ideation first generates many ideas with judgment deferred, then narrows them with criteria.	Critiquing or voting on ideas while they are still being generated.	A brainstorm that stalls because each idea is debated when it is raised.
5	Open, non-leading interview questions	Good research questions ask about past behavior and let the user describe it in their own words.	Asking yes/no or leading questions that suggest the answer the team hopes for.	"Wouldn't it be easier if...?" or "Would you use...?"
6	Empathy map quadrants	An empathy map sorts research into what the user says, thinks, does, and feels.	Mixing observed actions with quoted words, or treating it as a feature list.	A research note that must be placed in one quadrant.
7	Inclusive and equitable design	Designing with and for users most often excluded (edge cases, disabilities, low access) tends to produce better experiences for everyone.	Designing for the "average user" and treating other groups as out of scope.	A group the product currently fails, framed as too small to matter.
8	Journey maps expose pain points	A user journey map lays out the steps, actions, and emotions a user goes through to reach a goal, revealing where the experience breaks down.	Confusing a journey map with a site map or a task list of the system's screens.	A need to see where, across a sequence of steps, users become frustrated.

Course scope (public description): Empathizing, Defining, and Ideating prepares students to incorporate empathy-driven, user-centered design thinking in design solutions. This course focuses on the importance of empathy, research, and interdisciplinary collaboration in equitable design solutions. Students gain experience ideating with diverse workgroups to define the needs of users and stakeholders through research, determine audience segments and personas, and apply ideation strategies to create impactful, and meaningful user experience design concepts.